

BIG DATA ANALYTICS

UNIT II

Big Data Generation and Acquisition

Enterprise, IoT, Internet, and Biomedical Data Sources • Collection, Transportation, and Pre-processing



What We Will Cover in Unit II

Part A — Big Data Generation



Where does big data actually come from? Enterprise systems, IoT devices, the Internet, biomedical instruments, and other specialised fields.

Part B — Big Data Acquisition



How raw, scattered data is gathered, moved, and prepared: Data Collection, Data Transportation, and Data Pre-processing.

Outcome



By the end of this unit you should be able to classify a real dataset by its generation source and describe the acquisition pipeline it would need.



A

Big Data Generation

Understanding the five major sources that continuously produce today's data flood.



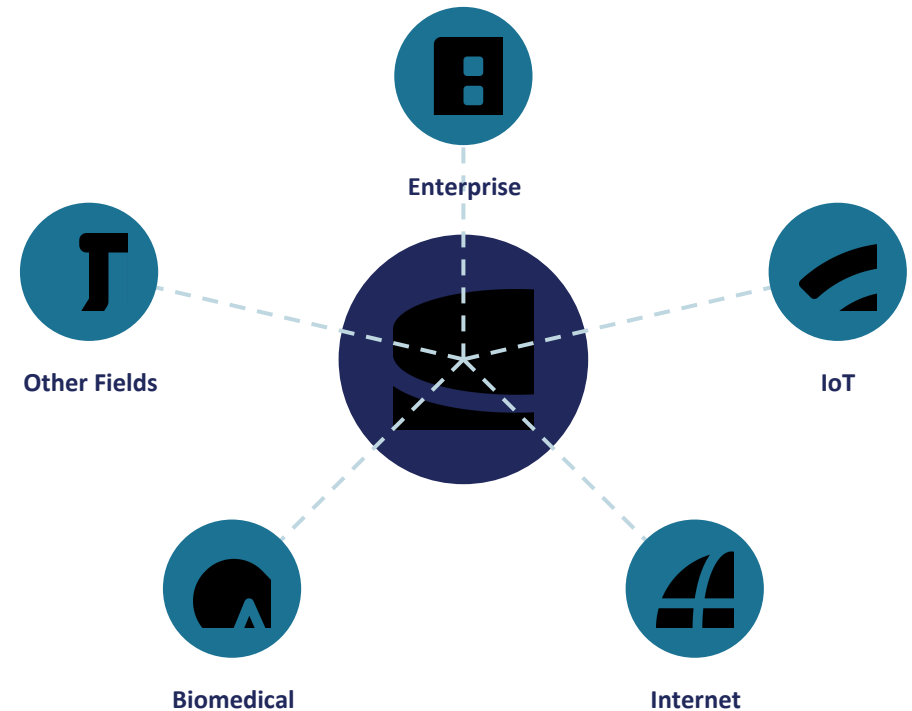
CONCEPT

What Is Big Data Generation?

Big Data Generation refers to the continuous, large-scale creation of data by many different activities, devices, and systems — long before any of it is stored or analysed. Every unit of data has a source, and understanding the source tells us its volume, velocity, variety, and veracity.

Five broad generation sources are studied in this unit:

- Enterprise Data
- IoT Data
- Internet Data
- Biomedical Data
- Data from Other Fields



Enterprise Data

Data generated inside organisations as part of their day-to-day operations.

Transactional Data



Records created every time a business event occurs — sales invoices, purchase orders, bank transactions, and inventory movements. Typically stored in relational databases (RDBMS) with strict schemas.

CRM & ERP Data



Customer Relationship Management systems store customer profiles, complaints, and interactions; Enterprise Resource Planning systems integrate finance, HR, supply chain, and manufacturing data across departments.

Operational Logs



System logs, audit trails, and employee activity records generated automatically by enterprise software to track 'what happened, when, and by whom' for compliance and troubleshooting.

Enterprise Data — Characteristics & Example



Key Characteristics

- Structured and semi-structured — fits neatly into tables, XML, or JSON
- Highly reliable and internally validated (high veracity)
- Moderate volume compared to IoT/Internet data, but very high business value
- Governed by strict access control, retention, and compliance policies



Classroom Example — Flipkart / FlipMart

During Diwali sale, FlipMart's enterprise systems generate:

- Order Management System → order IDs, SKU, quantity, price
- Payment gateway logs → UPI transaction references
- Warehouse ERP → stock levels across fulfilment centres
- CRM → seller ratings, customer complaint tickets

IoT Data

Data continuously streamed by the billions of connected devices in the Internet of Things.

Sensor Data



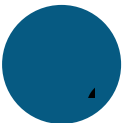
Temperature, humidity, pressure, motion, and location readings captured at regular intervals — for example, a smart electricity meter reporting usage every 15 minutes.

Wearable & Consumer Device Data



Fitness bands and smartwatches (e.g., Mi Band, Noise) continuously log heart rate, step count, and sleep patterns, uploading them to cloud dashboards.

Industrial & Vehicular Telemetry



Ola/Uber vehicles and factory machines emit GPS coordinates, engine diagnostics, and fuel levels in real time for fleet tracking and predictive maintenance.

IoT Data — Why It Is Different

Unlike enterprise data, IoT data is generated automatically, continuously, and at massive scale — often without any human triggering the event.

High Velocity



New readings every few seconds — a single smart city can generate millions of events per hour.

High Volume



Multiplied across millions of devices; a fleet of 10,000 sensors easily crosses terabytes per day.

Streaming Nature



Data must often be processed in real time (stream processing), not just stored for later batch analysis.

Variable Reliability



Devices can go offline, drift out of calibration, or transmit noisy readings, lowering veracity.

Internet Data

Data generated whenever people and machines interact with the web.

Social Media Data



Posts, likes, shares, comments, and hashtags on platforms like X (Twitter), Instagram, and Facebook — rich in sentiment but unstructured (text, images, video).

Search & Clickstream Data



Every search query, click, and page visit is logged — for example, Google search logs or a website's clickstream showing the exact path a visitor took.

Streaming & Web Traffic Data



Video/audio streaming platforms like Hotstar and Spotify generate logs of what is watched, for how long, and where users pause, buffer, or drop off.

Internet Data — Types at a Glance

Comparing the major categories of Internet-generated data.

Category	Typical Content	Structure	Example
Social Media	Posts, comments, hashtags, reactions	Unstructured	Trending hashtag analysis on X during IPL
Search Logs	Queries, click position, dwell time	Semi-structured	Google Trends showing festival-season searches
Web Traffic / Clickstream	Page views, session paths, bounce rate	Semi-structured	E-commerce site tracking cart abandonment
Streaming Logs	Play, pause, buffer, quality-switch events	Semi-structured	Hotstar logs during a cricket final

Biomedical Data

Data generated by healthcare systems, medical devices, and research laboratories.

Electronic Health Records (EHR)



Digitised patient records covering diagnoses, prescriptions, and visit history — for instance, hospital management systems linked to a patient's Ayushman Bharat Health ID.

Genomic & Omics Data



DNA sequencing produces enormous datasets (gigabytes per genome) used in personalised medicine and disease-risk prediction.

Medical Imaging & Device Data



MRI, CT, and X-ray scans, plus continuous readings from ICU monitors (heart rate, oxygen saturation) — high-volume, high-resolution, and time-critical.

Biomedical Data — Why It Needs Special Care

Highly Sensitive



Contains personally identifiable and health-related information; governed by strict privacy norms (e.g., HIPAA internationally, DISHA/health-data rules in India).

Very High Volume



A single genome (~200 GB raw) or one hospital's daily imaging output can rival an entire enterprise's yearly transactional data.

Multi-Modal Variety



Mixes structured records, free-text doctor notes, images, and time-series signals — all describing the same patient.

Life-Critical Veracity



Errors or delays in acquisition can directly impact patient safety, so accuracy and timeliness are non-negotiable.

Data Generation from Other Fields

Beyond enterprise, IoT, Internet, and biomedical sources, many specialised domains generate their own big data.

Scientific & Astronomical Research



Telescopes, particle accelerators, and climate sensors (e.g., ISRO satellites, weather radars) generate petabyte-scale observational data continuously.

Government & Public Sector Data



Census records, Aadhaar enrolments, CoWIN vaccination logs, and GST filings generate structured, nation-scale datasets used for policy-making.

Financial Markets & Logistics



Stock exchanges (NSE/BSE) generate microsecond-level tick data, while logistics networks (India Post, delivery fleets) generate continuous tracking data.

Comparing All Five Generation Sources

A single view to help you classify any dataset you encounter.

Source	Primary Driver	Structure	Volume/Velocity
Enterprise Data	Business transactions	Structured	Moderate volume, steady velocity
IoT Data	Connected devices/sensors	Semi-structured	High volume, very high velocity
Internet Data	Human/web interaction	Unstructured	Very high volume, bursty velocity
Biomedical Data	Healthcare & research	Multi-modal	Very high volume, mixed velocity
Other Fields	Science, government, finance	Structured/complex	Domain-dependent, often petabyte-scale

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B

Big Data Acquisition

How scattered raw data becomes a clean, usable input for analytics: Collection, Transportation, and Pre-processing.



The Big Data Acquisition Pipeline

Once data is generated, it must be acquired before it can be analysed — a three-stage pipeline.

1

Data Collection

Gathering raw data from its point of generation using logs, sensors, crawlers, and dedicated acquisition tools.



2

Data Transportation

Moving the collected data — often across networks and locations — to storage or processing systems, reliably and efficiently.



3

Data Pre-processing

Cleaning, integrating, transforming, and reducing raw data so it becomes accurate, consistent, and analysis-ready.

Data Collection

The process of gathering raw data from where it is generated using specific methods and tools suited to the source.

Log File Collection



Automatically capturing system-generated log entries — web server logs, application logs, or transaction logs — which record events with timestamps.

Sensing / Sensor-Based Collection



Physical sensors (temperature, GPS, RFID readers) continuously sample the environment and push readings to a collection endpoint.

Web Crawling / Scraping



Automated crawler programs (like the bots search engines use) systematically browse and extract content from web pages at scale.

Data Collection Methods Compared

Choosing the right collection method depends on the data source.

Method	Best Suited For	Tooling Example	Challenge
Log File Collection	Enterprise systems, servers	Log agents, Flume, Logstash	Log volume and format inconsistency
Sensor / RFID Collection	IoT, logistics, warehouses	MQTT brokers, edge gateways	Device connectivity and power limits
Web Crawling	Internet data, price/news tracking	Scrapy, custom crawlers	Dynamic pages, robots.txt, rate limits
API-Based Collection	Social media, payment platforms	REST/GraphQL APIs, webhooks	Rate limits, authentication, quota costs

A Closer Look: Crawlers and Sensor Networks



Web Crawlers

- Start from a set of 'seed' URLs and follow hyperlinks outward
- Respect robots.txt rules that tell crawlers which pages they may visit
- Store fetched pages in a queue for parsing and indexing
- Used by search engines, price-comparison sites, and news aggregators



Sensor Networks

- Many small sensor nodes sample a physical quantity (e.g., temperature)
- Nodes forward readings to a gateway using low-power protocols (Zigbee, LoRa, MQTT)
- Gateway aggregates and forwards data onward to cloud storage
- Used in smart agriculture, smart cities, and industrial monitoring

Data Transportation

Once collected, data must travel — often across long distances and multiple networks — to reach storage and processing systems.

Why Transportation Is Needed



Data is rarely generated where it is analysed. Sensor data from a factory floor must reach a central data centre or cloud region for processing.

Inter-DC and Intra-DC Transport



Inter-data-centre transport moves data between geographically separated centres; intra-data-centre transport moves it between servers within the same facility.

Physical vs. Network Transport



Small/medium data typically travels over IP networks; extremely large one-time transfers sometimes use physical transport (shipping hard-disk appliances) when it is faster than the network.

Challenges in Data Transportation

Bandwidth Limitations



Networks have a finite capacity; transporting IoT-scale or video data can saturate links and slow down other traffic.

Security & Privacy in Transit



Data can be intercepted while moving, so encryption (TLS/VPN) is essential — especially for biomedical or financial data.

Latency vs. Throughput Trade-off



Real-time applications (fraud detection) need low latency; bulk transfers (nightly backups) optimise for high throughput instead.

Reliability & Fault Tolerance



Network failures must not cause data loss — retry mechanisms, buffering, and acknowledgements are built into transport pipelines.

Improving Data Transportation



Network-Level Approaches

- Upgrading and re-architecting data-centre network topologies for higher bandwidth
- Software-Defined Networking (SDN) to dynamically route traffic around congestion
- Data compression before transport to shrink the volume travelling over the wire



Operational Approaches

- Edge computing — pre-processing data near its source so only relevant data is transported
- Scheduling large transfers during off-peak hours to avoid contention
- Using dedicated high-throughput links for known heavy transfer routes

Data Pre-processing — Why It Is Needed

Raw, freshly collected data is almost never ready for analysis. It typically arrives incomplete, inconsistent, noisy, or duplicated — pre-processing fixes this before any model or report can trust the data.

Incompleteness



Missing values — a sensor drops readings, or a customer leaves a form field blank.

Noise



Random errors or outliers — a mis-calibrated sensor reporting an impossible temperature.

Inconsistency



The same fact recorded differently across systems — 'Tamil Nadu' vs. 'TN' vs. 'T.Nadu'.

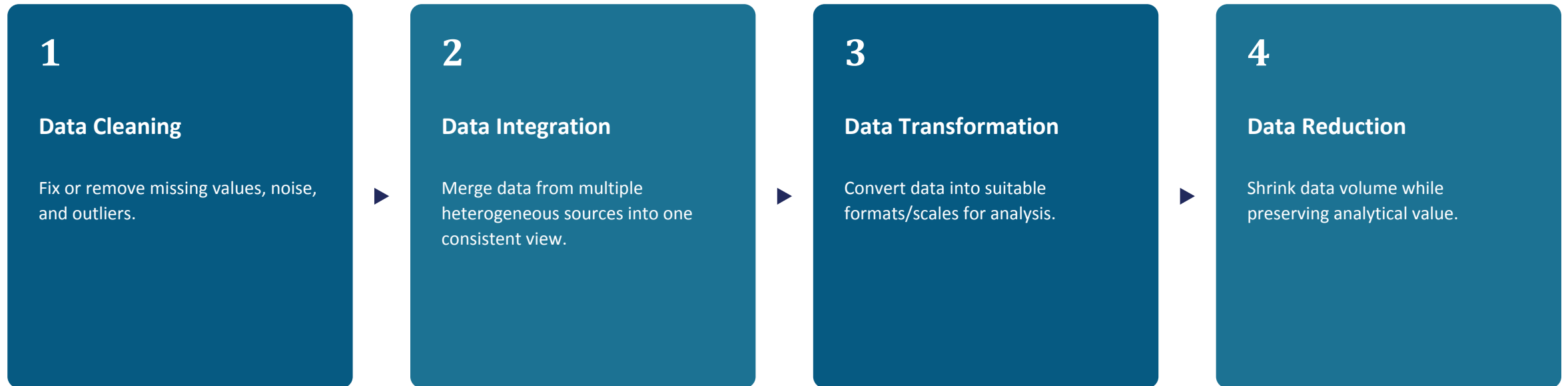
Redundancy



The same record duplicated across multiple source systems, inflating storage and skewing results.

The Four Stages of Data Pre-processing

Each stage tackles a specific data-quality problem, usually in this order.



Data Cleaning

The process of detecting and correcting (or removing) inaccurate, incomplete, or irrelevant records.

Handling Missing Values



Options include deleting incomplete records, filling gaps with the mean/median/mode, or predicting the missing value from related fields.

Handling Noisy Data



Techniques such as binning (grouping nearby values), regression smoothing, and outlier detection reduce random variation and errors.

Removing Duplicates & Irrelevant Data



Identifying records that refer to the same real-world entity and merging or removing them; dropping fields that add no analytical value.

Data Integration

Combining data from multiple sources — each possibly with a different schema and format — into one unified, coherent dataset.

Schema Integration



Matching corresponding fields across sources, e.g., recognising that 'Cust_ID' in one system and 'CustomerCode' in another refer to the same entity.

Entity Resolution



Detecting that two records — perhaps with slightly different spellings — describe the same real-world customer, product, or transaction.

Handling Redundancy & Conflicts



When the same attribute has different values across sources (e.g., two different phone numbers for a customer), rules decide which value to trust.

Data Transformation

Converting cleaned, integrated data into a format and scale that suits the intended analysis or model.

Normalisation / Scaling



Rescaling numeric attributes (e.g., Min-Max or Standard Scaling) so that no single feature dominates a model purely due to its unit or range.

Aggregation



Summarising fine-grained records into higher-level summaries — e.g., converting per-transaction UPI logs into daily total spend.

Discretisation & Encoding



Converting continuous values into categories (e.g., age → age-group), or categorical labels into numeric codes for machine learning models.

Data Reduction

Reducing the volume of data while preserving (as much as possible) the same analytical insights.

Dimensionality Reduction



Reducing the number of attributes/features analysed (e.g., Principal Component Analysis) while keeping most of the meaningful variation.

Numerosity Reduction



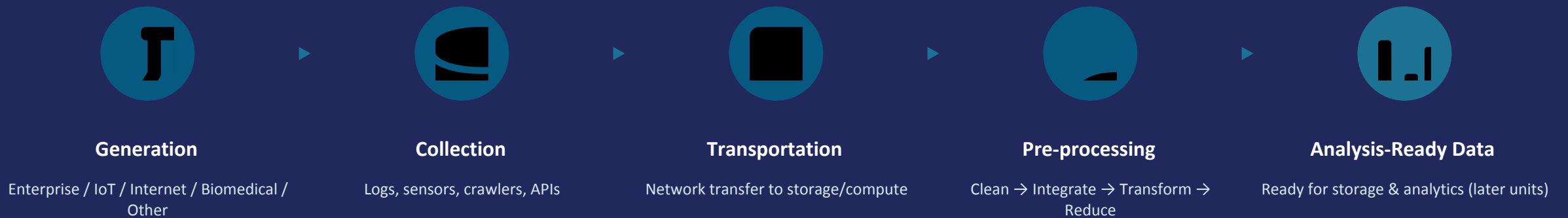
Reducing the number of records through sampling or building a smaller representative model of the data (e.g., histograms, clustering).

Data Compression



Encoding data in a smaller footprint (lossless or lossy) to reduce storage and transport cost without changing its underlying meaning.

End-to-End: From Generation to Analysis-Ready Data



Everything after this pipeline — storage, querying, and analytics — depends on how well generation and acquisition were handled here in Unit II.

Tracing a UPI Payment Through the Pipeline

Every time a customer pays a shopkeeper using UPI (say, on a Diwali shopping trip), the entire generation-to-preprocessing pipeline runs in seconds.



Generation

The UPI app (Internet/Enterprise data) generates a transaction record the instant the customer confirms payment.



Collection

The bank's and NPCI's transaction logs capture the event with a timestamp, amount, and unique reference ID.



Transportation

The transaction record is transported securely over encrypted networks between the payer bank, NPCI switch, and payee bank.



Pre-processing

Duplicate retries are removed, currency fields are standardised, and the record is merged with the merchant's sales data for reconciliation.

Key Terms Glossary

Quick definitions for revision before the internal assessment.

Term	One-Line Meaning
Enterprise Data	Data generated by an organisation's internal business systems (ERP, CRM)
IoT Data	Data streamed automatically by connected sensors and devices
Clickstream	The recorded sequence of pages/links a website visitor clicks through
Data Collection	Gathering raw data at the point where it is generated
Data Transportation	Moving collected data across networks to storage/processing systems
Data Cleaning	Detecting and fixing missing, noisy, or duplicate data
Data Integration	Combining data from multiple heterogeneous sources into one view
Data Reduction	Shrinking data volume while preserving analytical value

Unit II — Key Takeaways

Five generation sources



Enterprise, IoT, Internet, Biomedical, and Other Fields each produce data with distinct structure, volume, and velocity.

Three acquisition stages



Data Collection, Data Transportation, and Data Pre-processing turn scattered raw data into an analysis-ready dataset.

Pre-processing has four pillars



Cleaning, Integration, Transformation, and Reduction — each solving a specific data-quality problem.

Quality in, quality out



The reliability of every later stage of Big Data Analytics depends entirely on how well generation and acquisition were handled.



Questions & Discussion

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